

Programming with C

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May 10, 2026



Abstract

This is an early draft on a book on Programming with C.

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Preface

This is a book on programming with C.

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[book pdf]



[course website]

<https://thomasweise.github.io/programmingWithC>

This book was built using the following software:

```
1 Alpine Linux 3.23.3
2 Linux 6.17.0-1010-azure x86_64
3
4 python: 3.12.13
5 gcc --std=c23: (Alpine 15.2.0) 15.2.0 x86_64-alpine-linux-musl
6
7 texgit_py: 0.9.11
8 texgit_tex: 0.9.7
9 pycommons: 0.8.96
10 pdflatex: pdfTeX 3.141592653-2.6-1.40.29 (TeX Live 2026)
11 biber: 2.21
12 makeglossaries: 4.7 (2025-05-14)
13 ghostscript: 10.06.0 (2025-09-09)
14
15 date: 2026-05-10 06:33:01 +0000
```

Chapter 1

Introduction

Listing 1.1: Our first C program, which just prints *Hello World!* and exits. (stored in file `hello_world.c`; output in Listing 1.2)

```
1  /* Hello world program */
2
3  #include <stdio.h>
4
5  int main(void) {
6      printf("Hello, World!\n");
7      return 0;
8  }
```

↓ compile and run (if compilation succeeds) ↓

Listing 1.2: The standard output stream (stdout) of the program `hello_world.c` given in Listing 1.1.

```
1  $ gcc -Wall -Wextra -std=c23 -pedantic -o hello_world hello_world.c
2  # gcc 15.2.0 succeeded with exit code 0.
3  $ ./hello_world
4  Hello, World!
5  # ./hello_world succeeded with exit code 0.
```

Backmatter

Glossary

Bash is a shell used under **Ubuntu Linux**, i.e., the program that “runs” in the **terminal** and interprets your commands, allowing you to start and interact with other programs [3, 12, 19]. Learn more at <https://www.gnu.org/software/bash>.

C is a programming language, which is very successful in system programming situations [5, 13, 17].

IT information technology

Linux is the leading open source operating system, i.e., a free alternative to **Microsoft Windows** [1, 6, 14–16]. We recommend using it for this course, for software development, and for research. Learn more at <https://www.linux.org>. Its variant **Ubuntu** is particularly easy to use and install.

Microsoft Windows is a commercial proprietary operating system [2]. It is widely spread, but we recommend using a **Linux** variant such as **Ubuntu** for software development and for our course. Learn more at <https://www.microsoft.com/windows>.

Python The Python programming language [8, 10, 11, 18], i.e., what you will learn about in our book [18]. Learn more at <https://python.org>.

stderr The *standard error stream* is one of the three pre-defined streams of a console process (together with the **standard input stream** (**stdin**) and the **stdout**) [9]. It is the text stream to which the process writes information about errors and exceptions. If an uncaught **Exception** is raised in **Python** and the program terminates, then this information is written to **standard error stream** (**stderr**). If you run a program in a **terminal**, then the text that a process writes to its **stderr** appears in the console.

stdin The *standard input stream* is one of the three pre-defined streams of a console process (together with the **stdout** and the **stderr**) [9]. It is the text stream from which the process reads its input text, if any. The Python instruction **input** reads from this stream. If you run a program in a **terminal**, then the text that you type into the terminal while the process is running appears in this stream.

stdout The *standard output stream* is one of the three pre-defined streams of a console process (together with the **stdin** and the **stderr**) [9]. It is the text stream to which the process writes its normal output. The **print** instruction of Python writes text to this stream. If you run a program in a **terminal**, then the text that a process writes to its **stdout** appears in the console.

terminal A terminal is a text-based window where you can enter commands and execute them [1, 4]. Knowing what a terminal is and how to use it is very essential in any programming- or system administration-related task. If you want to open a terminal under **Microsoft Windows**, you can press **Win**+**R**, type in **cmd**, and hit **Enter**. Under **Ubuntu Linux**, **Ctrl**+**Alt**+**T** opens a terminal, which then runs a **Bash** shell inside.

Ubuntu is a variant of the open source operating system **Linux** [4, 7]. We recommend that you use this operating system to follow this class, for software development, and for research. Learn more at <https://ubuntu.com>. If you are in China, you can download it from <https://mirrors.ustc.edu.cn/ubuntu-releases>.

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